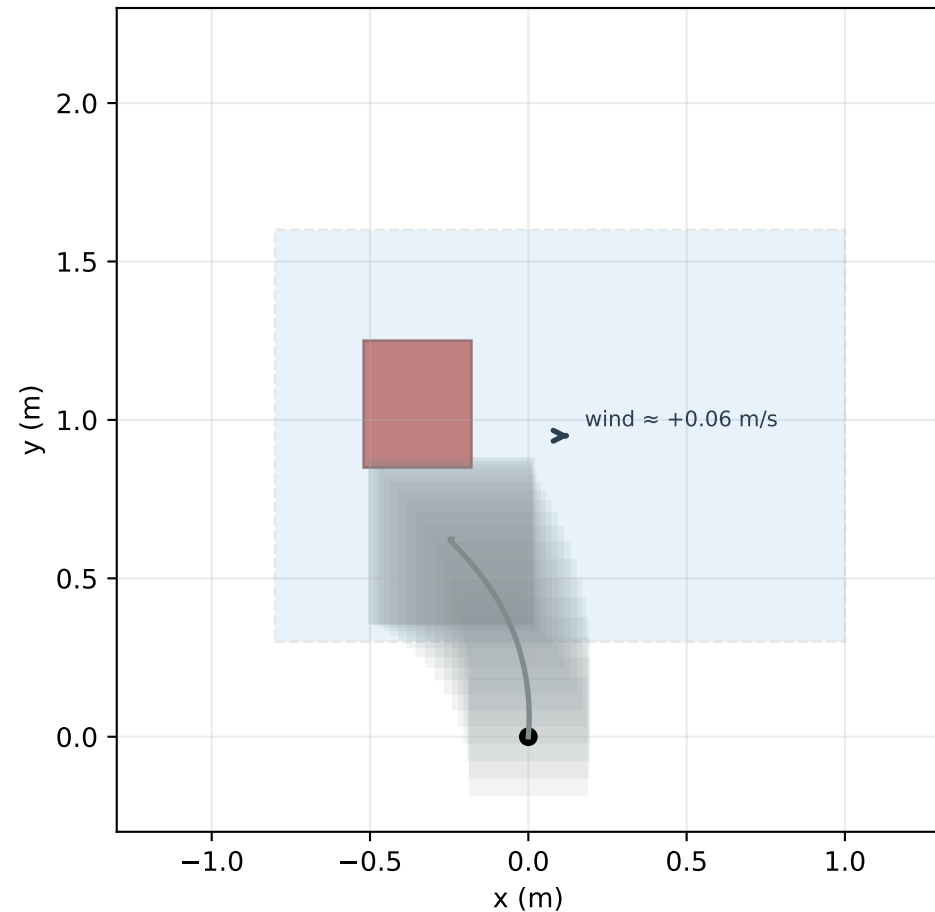
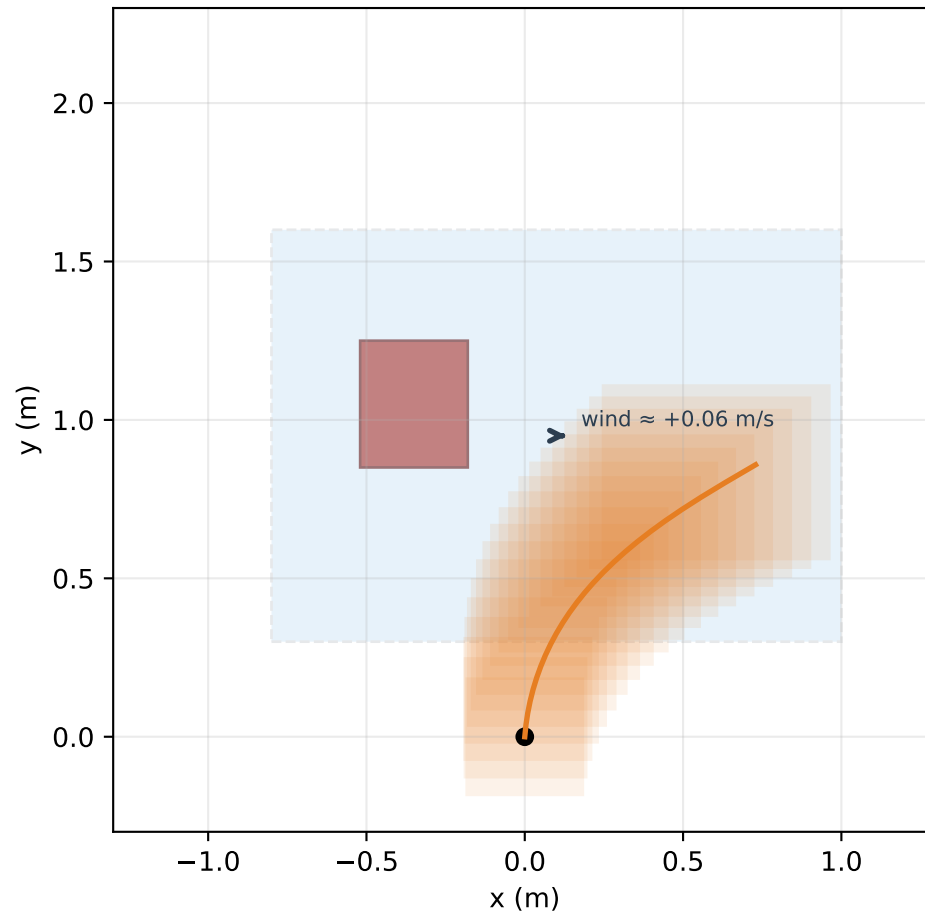


Planner-time disturbance awareness, scenario = wind
each panel = world-space tube at the planner-picked k

goal-only (no MMR)
 $k = (1.0, 0.28)$, hard $m_{dir} = -0.032$, $m_{sym} = -0.032 \rightarrow$ UNSAFE



goal + symmetric MMR constraint
 $k = (-1.0, -1.0)$, hard $m_{dir} = +0.167$, $m_{sym} = +0.115 \rightarrow$ safe



goal + directional MMR constraint
 $k = (-0.85, -1.0)$, hard $m_{dir} = +0.145$, $m_{sym} = +0.089 \rightarrow$ safe

